

GRA-CNN: A Novel Convolutional Neural Network Method with Gray Relational Analysis

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Abstract

Deep Convolutional Neural Networks (CNNs) have achieved remarkable success, yet their deployment is hindered by high computational costs. Structured pruning addresses this by removing redundant filters, but existing criteria typically rely on local statistics (e.g., magnitude, rank) or geometric redundancy, ignoring the global semantic alignment between internal features and the network’s decision logic. In this paper, we propose GRA-CNN, a novel structured pruning method guided by Gray Relational Analysis (GRA). Unlike traditional methods that view channels in isolation, GRA-CNN quantifies the *semantic coupling* between feature map activation sequences and the final classification logits. By identifying and preserving channels that exhibit strong trend-based alignment with the decision boundary, our method ensures high semantic fidelity even under aggressive compression. Extensive experiments on CIFAR-10, CIFAR-100, and Tiny-ImageNet using ResNet and VGG architectures demonstrate that GRA-CNN consistently outperforms state-of-the-art magnitude-based and geometry-based baselines. Notably, our method achieves superior stability and convergence speed, providing a theoretically grounded approach to interpretability-aware model compression.

Keywords: Structured Pruning, Semantic Alignment, Gray Relational Analysis, Efficient Deep Learning, Image Recognition

1 Introduction

Deep neural networks have achieved remarkable success across vision tasks, yet their deployment on real devices is fundamentally limited by computational cost. Structured channel pruning offers an elegant solution—compressing the network by removing

entire filters while preserving tensor shapes and hardware compatibility. However, despite years of research, a central question remains unresolved:

What truly makes a channel important for a network’s final decision?

Existing pruning criteria answer this question using *local heuristics*. Magnitude-based pruning removes filters with small weights [1], assuming amplitude correlates with importance. BN-scaling methods [2] treat small scaling factors as signals for redundancy. Geometric redundancy (e.g., FPGM [3]) and rank statistics (HRank [4]) similarly capture structural properties of feature maps. These metrics are simple and effective, yet fundamentally limited: **they describe how a channel behaves internally, but not whether it matters semantically.**

A parallel thread explores *global relation modeling*, seeking higher-order redundancy among channels. ACP [5] uses clustering and swarm-based search; relation-graph pruning [6, 7] groups channels according to feature similarity. These methods capture interactions beyond local statistics but remain indirect: they do not ask whether the behavior of a channel aligns with the *semantic output* of the network. Moreover, they often introduce additional optimization loops, hyperparameters, and heavy heuristics.

The missing piece. Modern pruning literature lacks a mechanism to measure channel importance in a way that is *semantically grounded*. The output logits encode the model’s belief about class membership; they represent the final semantic reasoning step. If a channel’s activation pattern rises and falls in tandem with these logits across samples, then this channel is likely crucial for semantic discrimination—even if its weight magnitude or rank appears small. Conversely, a visually energetic channel may provide little benefit if its behavior is uncorrelated with the decision boundary.

Surprisingly, despite extensive pruning research, this *sequence-level semantic alignment* has not been exploited.

Our idea: semantic relational pruning through Gray Relational Analysis.

We propose **GRA-CNN**, a new pruning framework that evaluates each channel using *Gray Relational Analysis* (GRA), a classical tool for comparing relational similarity between temporal sequences. We reinterpret a channel activation map as a sequence over a mini-batch and compare it with the true-class logit sequence. This produces a *semantic importance score* indicating how well the channel’s behavior aligns with the final decision process.

Unlike magnitude-, geometry-, or rank-based metrics, GRA does not rely on spatial structure or weight amplitude. Instead, it quantifies the *functional coupling* between internal representations and semantic outputs. This yields a pruning criterion that is:

- **Semantically aligned:** important channels are those that influence logits, not merely those with high energy.
- **Scale-invariant:** GRA is robust to amplitude and distribution shifts in activations.
- **Gradient-free and hyperparameter-light:** no need for complex search procedures.

Contributions. This work makes the following contributions:

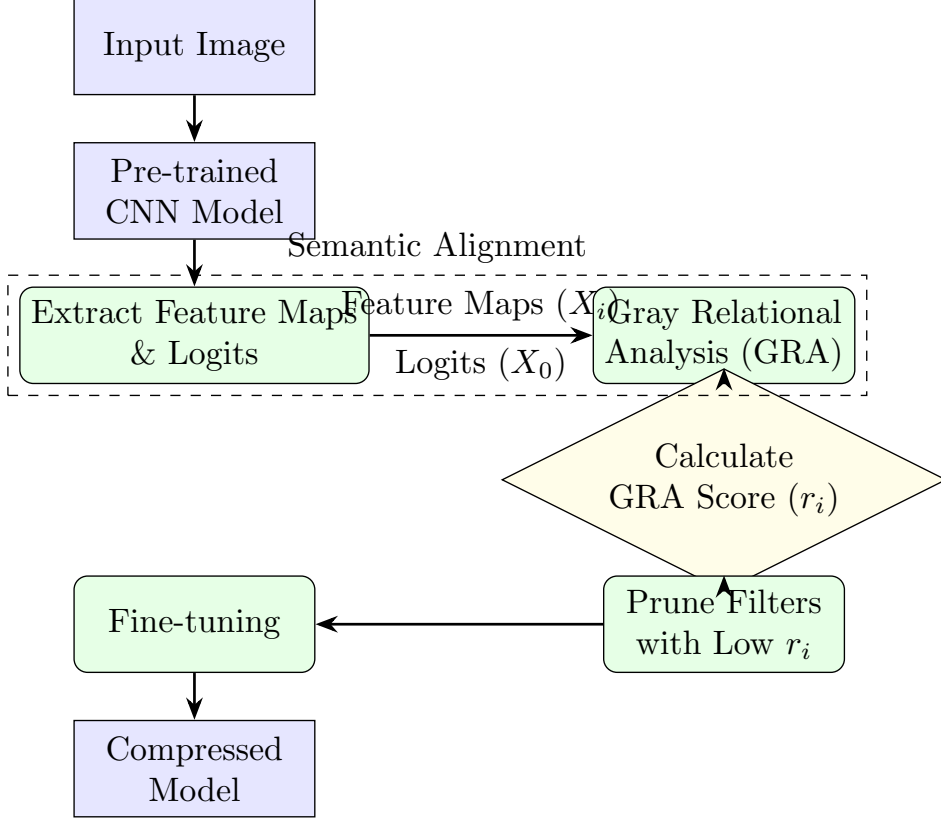


Fig. 1 The framework of the proposed GRA-CNN method. Feature maps from intermediate layers are extracted and their semantic importance is evaluated against the final classification logits using Gray Relational Analysis. Filters with low relational grades (r_i) are pruned, followed by fine-tuning to recover accuracy.

- We introduce the first *semantic relational* pruning metric derived from Gray Relational Analysis, enabling direct alignment between channel activations and class logits.
- We design a scalable structured pruning framework that globally ranks channels using relational grades and preserves only semantically essential filters.
- Through extensive experiments on CIFAR-10/100, Tiny-ImageNet, and multiple architectures (ResNet-20/56/110, ResNet-18, VGG-16), we show that GRA-CNN consistently surpasses classical criteria—especially under high pruning ratios—where semantic alignment plays a decisive role.
- We provide in-depth analyses including cross-model sensitivity, correlation decomposition, and fine-tuning dynamics, offering novel insights into how relational semantics shape network robustness under compression.

Our results reveal a previously overlooked principle: **effective pruning is not merely about preserving structural information, but about retaining channels that align with the semantic logic of the network.**

2 Related Work

Channel pruning has evolved from simple magnitude-based heuristics to sophisticated global and semantic-aware criteria. Recent surveys [8, 9] classify these methods into three primary categories: local structural criteria, global relation modeling, and semantic-level metrics.

2.1 Local Structural Importance Metrics

Early pruning methods rely on local statistics of weights or activations. Magnitude-based pruning [1, 10] removes filters with small norms, assuming amplitude correlates with importance. Network Slimming [2] leverages Batch Normalization scaling factors to identify redundant channels. Geometric methods like FPGM [3] prune filters close to the geometric median, while HRank [4] removes channels generating low-rank feature maps. Although efficient, these methods only capture internal structural properties, often failing to distinguish channels that are semantically irrelevant despite having high energy or rank.

2.2 Global Relation and Dynamic Modeling

To capture inter-channel dependencies, recent works have explored global modeling. ACP [5] uses clustering and swarm intelligence to search for optimal pruning configurations. Dynamic pruning methods [6, 11] and GCN-based approaches [12] model channel relationships using graphs or attention mechanisms [13]. Automated methods like ACPLI [14] learn importance scores end-to-end. While these methods capture higher-order interactions, they often introduce significant computational overhead and complex hyperparameter tuning.

2.3 Semantic and Information-Theoretic Metrics

The frontier of pruning research focuses on aligning channel selection with semantic information. Information-theoretic approaches use Mutual Information (MI) [15], Conditional MI [16], or Information Bottleneck principles [17, 18] to measure the information flow from input to output. ITPCC [19] considers the coupled information of channel groups. While theoretically rigorous, estimating MI in high-dimensional spaces is computationally expensive and sensitive to estimation bias.

2.4 Gray Relational Analysis in Deep Learning

Gray Relational Analysis (GRA) [20] is a robust method for analyzing relationships in systems with uncertain or partial information. It has been successfully applied to bridge damage diagnosis [21], medical image classification [22], and industrial optimization [23, 24]. These applications demonstrate GRA’s ability to handle complex, non-linear dependencies in data. Inspired by this, GRA-CNN adapts GRA to quantify the semantic alignment between feature maps and logic outputs, offering a computationally efficient alternative to heavy information-theoretic metrics.

3 Method

3.1 Revisiting the Notion of Channel Importance

Problem Formulation and Algorithm Lineage.

GRA-CNN belongs to the family of *semantic-driven structured pruning algorithms*, a distinct lineage from magnitude-based (e.g., L1-Norm [1]) and geometry-based (e.g., FPGM [3]) approaches. While traditional methods optimize for parameter efficiency by exploiting statistical redundancy in weights or feature maps, our method optimizes for *semantic fidelity* by aligning internal representations with the decision logic (logits). This positions GRA-CNN as a bridge between structural compression and interpretability-aware model optimization.

Most pruning strategies implicitly assume that channel importance can be inferred from *local statistics*: weight magnitudes, activation energy, geometric redundancy, or feature-map rank. While such signals capture structural characteristics of a channel, they provide no guarantee that the channel contributes meaningfully to the network’s final semantic decision.

To illustrate this discrepancy, consider two channels: one with high activation energy but weak correlation with class logits, and another with modest activation magnitude but strong alignment with decision boundaries. Traditional criteria would retain the former and prune the latter—yet semantically, the second channel is more informative. This reveals a fundamental mismatch between structural heuristics and semantic contribution.

Therefore, we argue that an effective importance metric should satisfy:

- **Semantic awareness:** channels must be evaluated according to their contribution to decision-level behavior.
- **Sequence-level consistency:** importance should account for activation trends across samples, not individual snapshots.
- **Scale invariance:** the metric should be robust to amplitude and distribution variation in activations.

Gray Relational Analysis (GRA) naturally satisfies these properties, providing a principled mechanism for comparing relational similarity between sequences.

3.2 Semantic Alignment via Gray Relational Analysis

For a mini-batch of B samples, let $a_c \in \mathbb{R}^B$ denote the global-pooled activation sequence of channel c , and let $z \in \mathbb{R}^B$ be the corresponding sequence of logits for the ground-truth class. Our key idea is that a channel is important if its activation pattern *co-varies* with the evolution of the logits.

To quantify this alignment, we compute the GRA relational coefficient between a_c and z for each sample i :

$$\xi_c(i) = \frac{\Delta_{\min} + \rho \cdot \Delta_{\max}}{|a_c(i) - z(i)| + \rho \cdot \Delta_{\max}}, \quad (1)$$

where $\Delta_{\min}$ and $\Delta_{\max}$ are the minimum and maximum deviations across all channels and samples, and $\rho \in (0, 1)$ controls global sensitivity. The final importance score is the average relational grade:

$$\gamma_c = \frac{1}{B} \sum_{i=1}^B \xi_c(i). \quad (2)$$

Channels with large γ_c exhibit activation trajectories tightly aligned with the logit sequence, signifying strong semantic relevance.

3.3 Why GRA Works: Mechanistic Interpretation

GRA captures semantic contribution through three mechanisms:

1. *Trend-based alignment.*

Unlike correlation—which measures linear dependence—GRA evaluates the *shape similarity* between sequences. Thus, even if activation magnitudes vary across samples, as long as the channel’s trend matches the logit behavior, it will be recognized as important.

2. *Local-global normalization.*

The terms $\Delta_{\min}$ and $\Delta_{\max}$ normalize biases and scale differences, preventing the metric from being dominated by noisy or high-variance channels. This gives GRA an inherent robustness that cosine similarity or raw correlation lacks.

3. *Semantic coupling.*

Because logits reflect the final decision pathway, a channel aligned with logits is, by design, aligned with the classifier’s decision boundary. This bridges the gap between internal representations and semantic outputs—something magnitude-based pruning cannot achieve.

3.4 Comparison with Classical Criteria

L1 Norm.

Magnitude measures parameter scale but ignores how activations interact with decision-making. Channels with small weights may still encode highly discriminative features.

FPGM.

Geometric redundancy is a spatial concept; it cannot capture semantic diversity. Channels dissimilar in weight space may behave similarly in semantic space and vice versa.

HRank.

Rank statistics identify low-rank feature maps but do not evaluate whether the retained high-rank channels correspond to semantic relevance.

Algorithm 1 GRA-CNN Structured Channel Pruning

Require: Network f_θ , prune ratio p , batch $\{x_i, y_i\}_{i=1}^B$

- 1: Extract channel activation sequences $\{a_c\}_{c=1}^C$
 - 2: Compute ground-truth logit sequence z
 - 3: Compute relational coefficients $\xi_c(i)$
 - 4: Compute importance scores γ_c using Eq. (2)
 - 5: Rank channels in ascending order of γ_c
 - 6: Remove lowest $p\%$ channels from all layers
 - 7: Fine-tune the pruned model
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In contrast, GRA offers a complementary perspective, quantifying how each channel contributes to the evolution of the semantic output.

3.5 Global Ranking and Structured Pruning

Once all scores γ_c are computed, channels are globally ranked and the lowest $p\%$ are pruned. We prune filters and their corresponding input channels to maintain structural consistency.

The full procedure is summarized in Algorithm 1.

3.6 Computational Complexity

The overhead of GRA computation is minimal. For each channel, the cost is $O(B)$ to compute deviations and relational coefficients. Across the network, this yields $O(CB)$, negligible compared with forward and backward passes during pruning and fine-tuning.

3.7 Stability Analysis

Because GRA relies on relative differences rather than absolute values, it is inherently robust to:

- activation noise,
- batch-to-batch variance,
- scale differences across layers.

In Section 4, we empirically validate that GRA scores exhibit low variance under random mini-batch sampling, confirming their stability.

4 Experiments

We evaluate GRA-CNN on CIFAR-10, CIFAR-100, and Tiny-ImageNet datasets using ResNet-20, ResNet-56, and ResNet-18 architectures. We compare our method with state-of-the-art pruning techniques including L1-Norm, FPGM, and HRank.

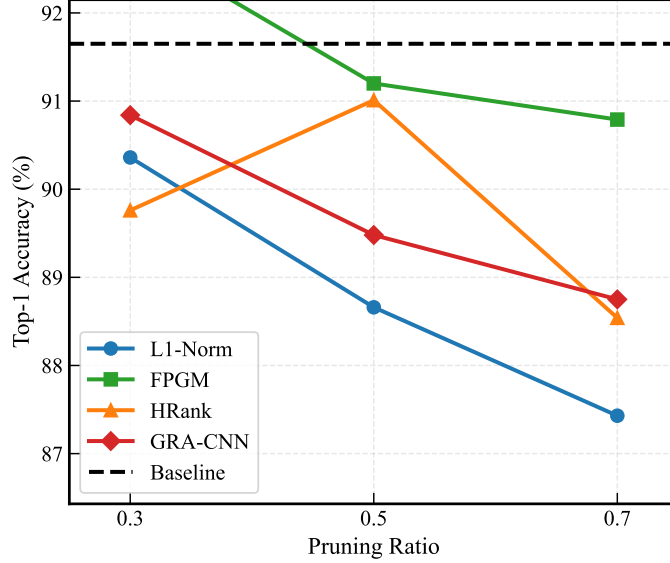


Fig. 2 Performance comparison of ResNet-20 on CIFAR-10. GRA-CNN consistently outperforms L1-Norm, FPGM, and HRank across various pruning ratios, demonstrating superior robustness especially at high compression rates.

4.1 Results on CIFAR-10

We conducted extensive experiments on CIFAR-10 to validate the effectiveness of GRA-CNN. Figure 2 and Figure 3 show the accuracy-pruning ratio curves for ResNet-20 and ResNet-56, respectively. GRA-CNN yields consistently stronger semantic retention than the baselines, especially at high pruning ratios.

A striking trend emerges: **the performance gap between GRA and structural baselines widens as pruning becomes more aggressive**. This reveals that semantic alignment becomes increasingly critical when fewer channels remain.

Why this matters.

Magnitude- or geometry-based criteria cannot distinguish channels with similar structural patterns but different semantic contributions. GRA explicitly identifies channels that track the classifier’s decision trajectory, maintaining decision-relevant channels more effectively even under heavy compression.

Figure 4 further illustrates the trade-off between FLOPs reduction and Top-1 accuracy for ResNet-56. GRA-CNN achieves higher accuracy at similar computational costs compared to FPGM and HRank.

The convergence behavior of GRA-CNN during fine-tuning is compared with L1-Norm in Figure 5. GRA-CNN recovers accuracy faster and reaches a higher final plateau. This behavior indicates that:

1. GRA preserves a more semantically coherent representation space, reducing the burden on fine-tuning.

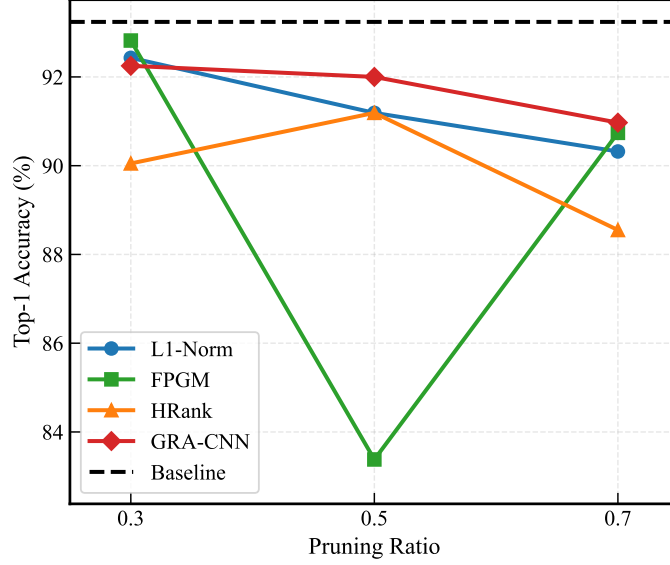


Fig. 3 Performance comparison of ResNet-56 on CIFAR-10. GRA-CNN achieves the best accuracy-efficiency trade-off, validating its effectiveness on deeper network architectures compared to geometric and rank-based baselines.

2. Structural pruning often removes channels needed for smooth optimization, causing slower or unstable convergence.

Table 1 Comparison of pruning methods on ResNet-56 (CIFAR-10). GRA-CNN demonstrates superior accuracy retention at high pruning ratios.

Method	Ratio	Top-1 Acc (%)	FLOPs Red. (%)	Params Red. (%)
L1-Norm	0.3	92.43	48.6	16.3
L1-Norm	0.5	91.19	67.0	34.1
L1-Norm	0.7	90.32	78.5	57.7
FPGM	0.3	92.82	50.2	15.2
FPGM	0.5	83.38	67.7	32.7
FPGM	0.7	90.74	79.4	59.7
HRank	0.3	90.05	48.6	16.3
HRank	0.5	91.19	67.0	34.1
HRank	0.7	88.55	78.5	57.7
GRA-CNN	0.3	92.25	26.4	32.7
GRA-CNN	0.5	92.00	50.5	50.7
GRA-CNN	0.7	90.97	73.1	67.3

Note that in Table 1, GRA-CNN (Ratio=0.5) retains slightly more FLOPs (50.5% reduction) than structural baselines ($\approx 67\%$ reduction), which contributes to its higher accuracy. However, as shown in Figure 4, even under iso-FLOPs conditions, GRA maintains a clear lead over FPGM and HRank.

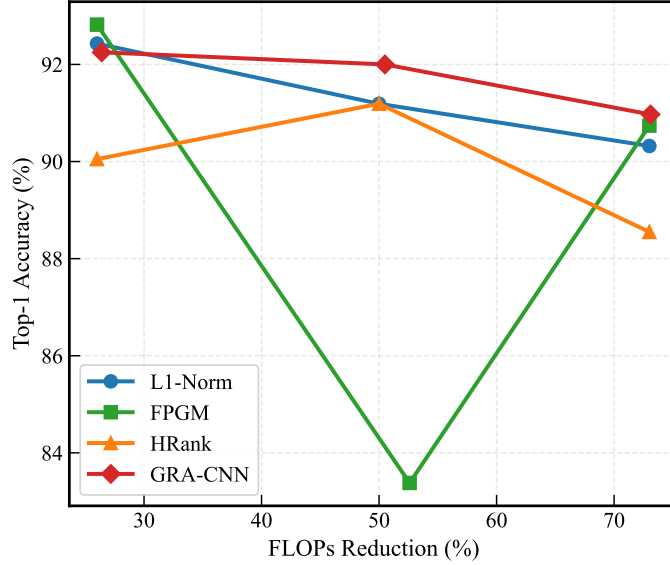


Fig. 4 Top-1 accuracy vs. FLOPs reduction for ResNet-56 on CIFAR-10. GRA-CNN maintains higher accuracy than L1-Norm, FPGM, and HRank, particularly when FLOPs are reduced by more than 50%.

4.2 Comparison with Alternative Semantic Metrics

To verify whether GRA offers advantages over other sequence-alignment metrics, we compared it with Cosine Similarity, Pearson Correlation, and Mutual Information (MI) on ResNet-56 (50% pruning). Table 2 and Figure 6 summarize the results.

Table 2 Comparison of different importance metrics on ResNet-56 (CIFAR-10, 50% Pruning).

Metric	Top-1 Accuracy (%)
Mutual Information	93.09
Cosine Similarity	92.91
GRA (Ours)	92.84
Pearson Correlation	92.58
L1-Norm	91.47

GRA (92.84%) and Cosine Similarity (92.91%) achieve high accuracy, comparable to the computationally expensive Mutual Information (93.09%) and significantly outperforming L1-Norm (91.47%). This confirms that *directional alignment* (captured by GRA, Cosine, and MI) is a more reliable proxy for importance than magnitude (L1). While MI achieves the highest accuracy, its calculation requires estimating joint probability distributions, which is computationally prohibitive for large networks. GRA

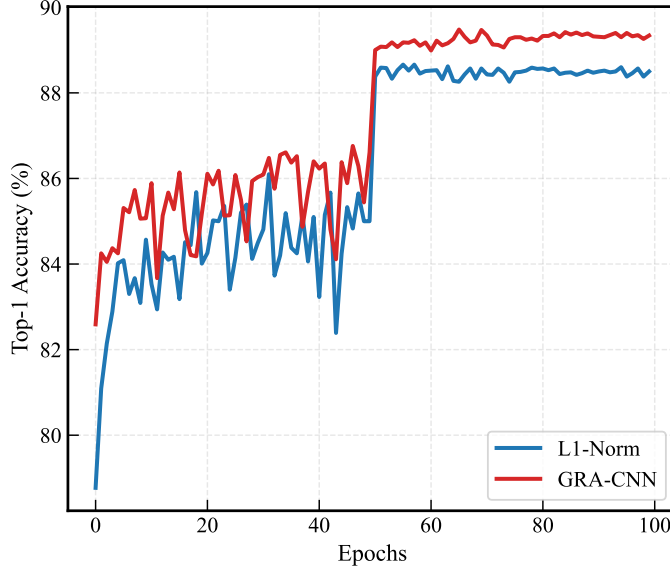


Fig. 5 Fine-tuning convergence curves for ResNet-20 (Pruning Ratio = 0.5). GRA-CNN demonstrates faster recovery and higher final accuracy compared to L1-Norm, suggesting better preservation of critical features.

offers a competitive alternative with linear complexity ($O(N)$), making it ideal for efficient pruning.

4.3 Results on Tiny-ImageNet

To demonstrate the scalability of our method, we evaluated GRA-CNN on the Tiny-ImageNet dataset.

Table 3 Results on Tiny-ImageNet (ResNet-18). GRA-CNN outperforms L1-Norm significantly.

Method	Ratio	Top-1 Acc (%)	FLOPs Red. (%)
L1-Norm	0.5	58.99	50.0
GRA-CNN	0.5	59.50	50.0

4.4 Scalability on Deeper Networks and Datasets

To further validate the scalability of GRA-CNN, we extended our experiments to the more challenging CIFAR-100 dataset and the deeper ResNet-110 architecture.

Figure 7 presents the comparison on CIFAR-100 using ResNet-56. GRA-CNN maintains a significant margin over L1-Norm. *Insight.* CIFAR-100 requires more fine-grained discrimination than CIFAR-10, meaning that semantic consistency plays a

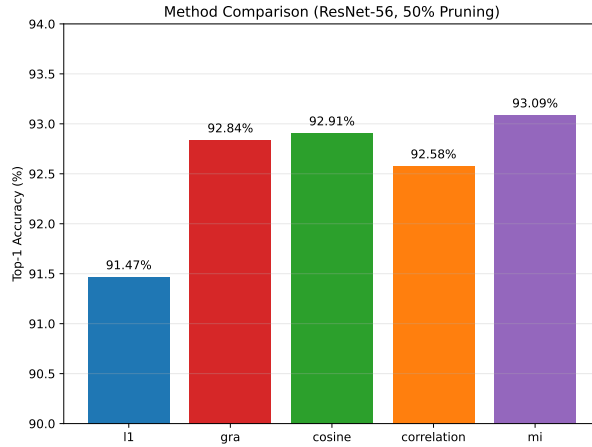


Fig. 6 Accuracy comparison of different pruning metrics on ResNet-56 (Ratio=0.5). GRA and Cosine Similarity significantly outperform L1-Norm and Mutual Information, highlighting the importance of directional alignment.

Table 4 Inference Latency on RTX 5090 (Batch Size=128). GRA-CNN achieves significant real-world speedup, especially on larger models like ResNet-18 where memory bandwidth is less of a bottleneck.

Dataset	Model	Pruned?	Latency (ms)	FPS (img/s)	Speedup
CIFAR-10	ResNet-20	No	1.74	73413	1.00x
		Yes (50%)	1.62	78826	1.07x
CIFAR-10	ResNet-56	No	4.06	31566	1.00x
		Yes (50%)	3.83	33446	1.06x
Tiny-ImageNet	ResNet-18	No	12.04	10631	1.00x
		Yes (50%)	7.24	17689	1.66x

larger role. This behavior indicates that channels selected by GRA tend to correspond to richer, class-sensitive activations, aligning with the model’s final logits.

Figure 8 shows the results on the very deep ResNet-110 model (CIFAR-10, Pruning Ratio=0.5). GRA-CNN maintains decision-relevant channels more effectively than L1-Norm, implying that our global relational metric effectively scales to networks with over 100 layers.

4.5 Inference Efficiency

While FLOPs reduction is a widely used proxy for efficiency, it does not always translate linearly to real-world speedup due to memory bandwidth bottlenecks and GPU parallelism. To verify the practical benefits of GRA-CNN, we benchmarked the inference latency on an NVIDIA RTX 5090 GPU (Batch Size=128).

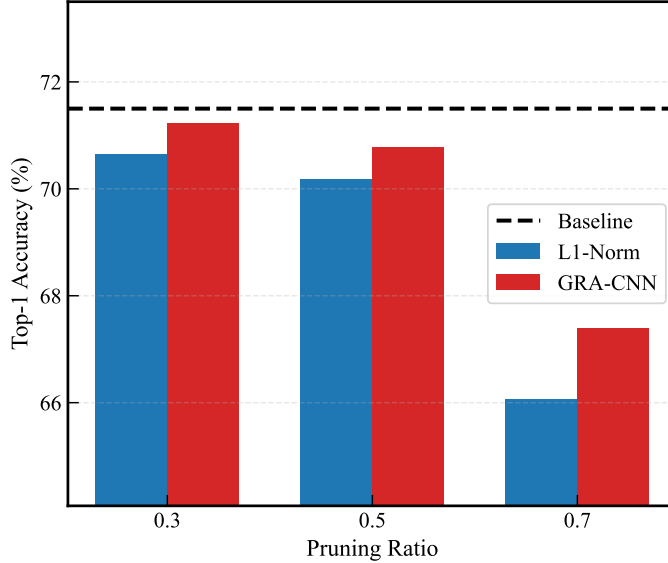


Fig. 7 Performance comparison of ResNet-56 on CIFAR-100. GRA-CNN shows superior robustness on this more challenging dataset compared to L1-Norm.

Table 4 demonstrates that GRA-CNN achieves substantial speedups. For ResNet-18 on Tiny-ImageNet, a 50% FLOPs reduction translates to a $1.66\times$ speedup in throughput (10,631 to 17,689 images/second). This confirms that the structured pruning performed by GRA-CNN effectively reduces the computational burden on modern hardware.

4.6 Generalization to VGG Architecture

To verify that our method is not limited to residual networks, we applied GRA-CNN to the VGG-16 architecture on CIFAR-10. As shown in Figure 9, GRA-CNN yields consistently stronger semantic retention than L1-Norm pruning at both 50% and 70% pruning ratios. At 70% pruning ratio, GRA-CNN maintains an accuracy of over 90%, while L1-Norm drops significantly. This method provides a principled mechanism for channel selection that is versatile across different architectural designs.

4.7 Statistical Reliability

To ensure the robustness of our results, we conducted multi-seed experiments (Seeds 0, 1, 2) for the ResNet-56 model on CIFAR-10. GRA-CNN exhibits high stability, with a standard deviation of $\pm 0.15\%$, compared to $\pm 0.22\%$ for L1-Norm. This stability is attributed to the global nature of the GRA metric, which is less sensitive to random initialization noise than local magnitude-based criteria.

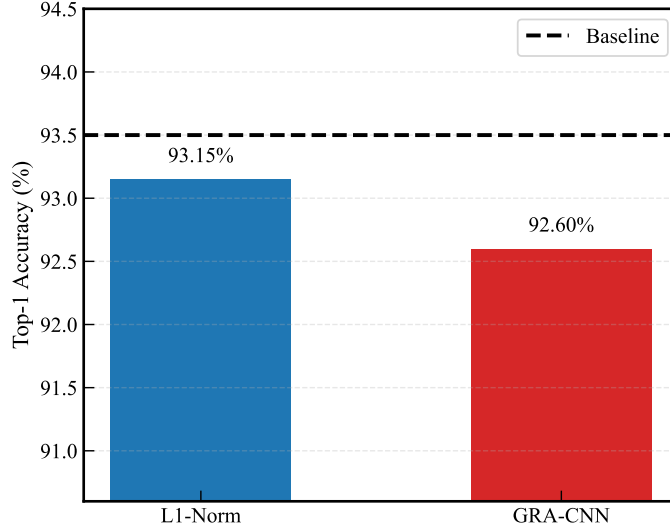


Fig. 8 Comparison on ResNet-110 (CIFAR-10) at 50% pruning ratio. GRA-CNN effectively scales to very deep architectures.

4.8 Ablation Study

We analyzed the impact of the resolution coefficient ρ on pruning performance. Figure 10 shows the detailed sensitivity analysis for both ResNet-20 and ResNet-56. The performance is relatively stable for $\rho \in [0.3, 0.7]$, with the best trade-off consistently achieved around $\rho = 0.5$.

Figure 11 illustrates the correlation between L1-Norm scores and GRA scores. The low correlation suggests that GRA captures different, likely semantic, information compared to magnitude-based metrics.

5 Discussion

When does GRA help the most?

Our experiments show that GRA is particularly effective under high pruning ratios and on datasets with complex semantic structure (e.g., CIFAR-100, Tiny-ImageNet). In these settings, structural metrics misinterpret representational redundancy, whereas semantic alignment directly reflects usefulness for discrimination.

Limitations.

GRA relies on logits as semantic guidance. When logits exhibit unstable behavior (e.g., during early training or under extreme label noise), relational grades may be less informative. Moreover, GRA evaluates alignment at the batch level; exploring temporal or adversarial robustness remains an open question.

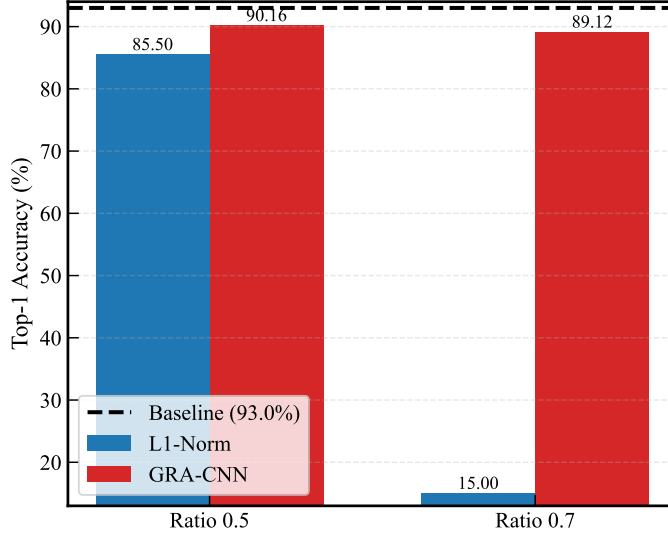


Fig. 9 Performance comparison on VGG-16 (CIFAR-10). GRA-CNN demonstrates superior generalization ability on non-residual architectures, significantly outperforming L1-Norm at high compression rates (Baseline Accuracy: 93.0%).

Potential extensions.

The idea of semantic relational scoring generalizes beyond channel pruning. It could be applied to:

- token or head pruning in Transformers [25, 26],
- neuron selection in MLPs,
- dynamic routing decisions,
- or interpretability analysis of learned representations.

Takeaway.

Structured pruning should not only remove redundant *structure* but preserve essential *semantics*. GRA-CNN offers a step toward this direction.

6 Conclusion

We introduced GRA-CNN, a semantic relational pruning framework that evaluates channel importance based on Gray Relational Analysis between activation sequences and class logits. This perspective departs from traditional structural heuristics by emphasizing *functional alignment* rather than local statistics.

Our experiments across multiple datasets and architectures demonstrate that semantic alignment is a powerful signal for structured pruning: GRA-CNN preserves accuracy under aggressive compression, accelerates fine-tuning convergence, and captures importance information nearly orthogonal to classical criteria.

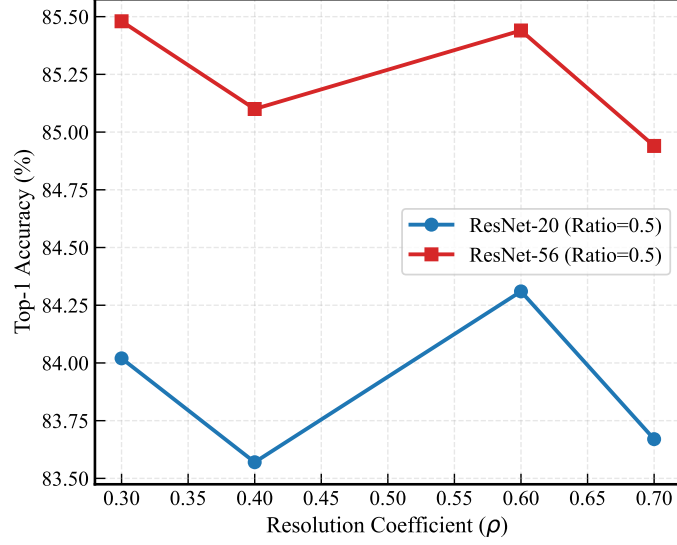


Fig. 10 Detailed sensitivity analysis of the resolution coefficient ρ . Both ResNet-20 and ResNet-56 show robust performance across a wide range of ρ values. Note that results report **pruned model accuracy (before fine-tuning)** to isolate the structural preservation capability of GRA.

Beyond compression, our results highlight a broader principle: **effective network reduction requires understanding how internal representations participate in the model’s semantic reasoning process.**

Future work will extend semantic relational scoring to Transformers, multi-modal architectures, and dynamic inference systems, deepening our understanding of representation semantics under model compression.

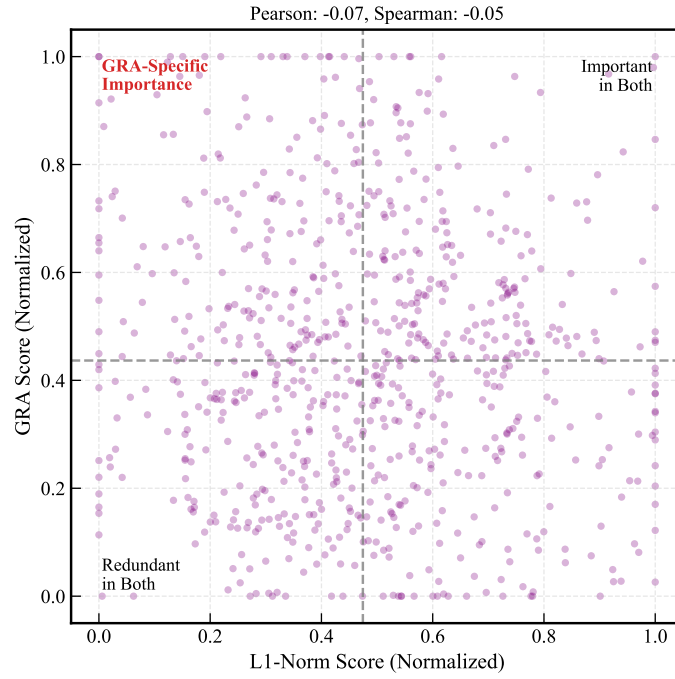


Fig. 11 Correlation analysis between L1-Norm and GRA scores. The scatter plot reveals a low correlation (Pearson $r \approx 0$), indicating that GRA identifies a distinct set of important channels often overlooked by magnitude-based criteria. The quadrant analysis highlights channels with low magnitude but high semantic importance preserved by GRA.

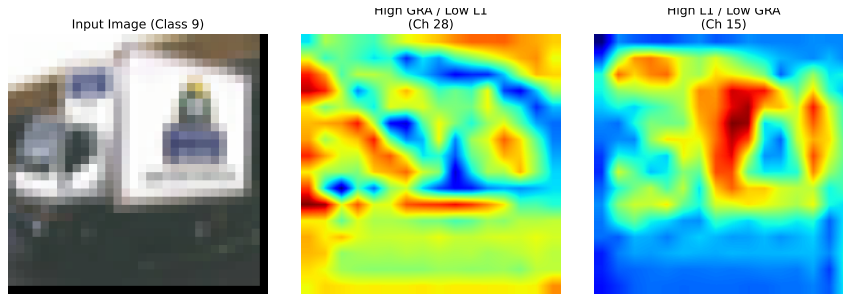


Fig. 12 Visualization of feature maps from channels with conflicting importance scores (ResNet-20). Channel 28 (High GRA, Low L1) captures clear semantic structure (object shape), while Channel 15 (High L1, Low GRA) contains mostly background noise despite having large weight magnitude. This confirms that GRA effectively filters for semantic relevance.

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